WORKSHEET

Chapter 5: Integrals of vector functions

1 Find the indefinite integral for each of the following.

a $\int (4t\mathbf{i} + t^2\mathbf{j})dt$

b $\int (2\sqrt{t}\mathbf{i} + 7t^3\mathbf{j})dt$

c $\int (9t^{\frac{3}{2}}\mathbf{i} - 8t^2\mathbf{j})dt$

d $\int \left[(6 - 2e^{-t})\mathbf{i} - (3t + 7)\mathbf{j}\right]dt$

e $\int \left[\sin(3t)\mathbf{i} + \cos(t)\mathbf{j}\right]dt$

f $\int \left[4\cos(5t)\mathbf{i} + 8e^{3t}\mathbf{j}\right]dt$

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2 In each of the following, find $\mathbf{r}(t)$ for the given conditions,

a $\dot{\mathbf{r}}(t) = \mathbf{i} + 3(t-1)\mathbf{j}$ and $\mathbf{r}(1) = \mathbf{i} + \mathbf{j}$

b $\dot{\mathbf{r}}(t) = 20\mathbf{i} + (15 - 10t)\mathbf{j}$ and $\mathbf{r}(0) = 4\mathbf{j}$

c $\dot{\mathbf{r}}(t) = -2\sin(t)\mathbf{i} + 3\cos(t)\mathbf{j}$ and $\mathbf{r}\left(\frac{\pi}{2}\right) = 5\mathbf{j}$

d $\dot{\mathbf{r}}(t) = e^{t}\mathbf{i} + e^{-t}\mathbf{j}$ and $\mathbf{r}(0) = \mathbf{i} + 2\mathbf{j}$

e $\dot{\mathbf{r}}(t) = \frac{1}{2}\mathbf{i} + 2t\mathbf{j}$ and $\mathbf{r}(0) = \mathbf{i} + \mathbf{j}$

f $\dot{\mathbf{r}}(t) = e^{t}\mathbf{i} + \frac{4}{9}e^{2t}\mathbf{j}$ and $\mathbf{r}(0) = 3\mathbf{i} + \frac{2}{9}\mathbf{j}$

3 In each of the following, find $\mathbf{r}(t)$ for the given conditions,

a $\ddot{\mathbf{r}}(t) = 3\mathbf{i}$, $\dot{\mathbf{r}}(0) = 5\mathbf{j}$ and $\mathbf{r}(0) = 2\mathbf{i} - 3\mathbf{j}$

b $\ddot{\mathbf{r}}(t) = 3\cos(t)\mathbf{i} - 2\sin(2t)\mathbf{j}$, $\dot{\mathbf{r}}(0) = 2\mathbf{j}$ and $\mathbf{r}(0) = -4\mathbf{i} + 4\mathbf{j}$

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c $\ddot{\mathbf{r}}(t) = 3e^{3t}\mathbf{i} - 5e^{-t}\mathbf{j}$, $\dot{\mathbf{r}}(0) = \mathbf{i} + 5\mathbf{j}$ and $\mathbf{r}(0) = \frac{5}{3}\mathbf{i} + 10\mathbf{j}$

d $\ddot{\mathbf{r}}(t) = 2\mathbf{j}$, $\dot{\mathbf{r}}(0) = \frac{1}{2}\mathbf{i}$ and $\mathbf{r}(0) = \mathbf{i} - 3\mathbf{j}$

4 Evaluate the following definite integrals.

a $\int_0^2 (4t\mathbf{i} - t\mathbf{j})dt$

b $\int_{-1}^1 (3t\mathbf{i} + t^2\mathbf{j})dt$

c $\int_0^{\frac{\pi}{2}} [\sin(t)\mathbf{i} - \cos(t)\mathbf{j}]dt$

d $\int_{\frac{-\pi}{6}}^{\frac{\pi}{6}} [\sin 3(t)\mathbf{i} - \cos(2t)\mathbf{j}]dt$

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e $\int_0^1 (4t\mathbf{i} - e^{5t}\mathbf{j})dt$

f $\int_0^1 (e^{-2t}\mathbf{i} + 2e^{t}\mathbf{j})dt$

5 Evaluate the definite integral $\int_0^2 (e^{4t}\mathbf{i} - \sqrt{t+1}\mathbf{j})dt$.

6 Evaluate $\int_0^{\frac{\pi}{2}} \left\{\left[\sin^2(t)\cos(t)\right]\mathbf{i} + \left[\sin(t)\cos^2(t)\right]\mathbf{j}\right\}dt$.

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Answers

1 **a** $2t^2\mathbf{i} + \frac{t^3}{3}\mathbf{j} + \mathbf{c}$

b $\frac{4\sqrt{t^3}}{3}\mathbf{i} - \frac{7t^4}{4}\mathbf{j} + \mathbf{c}$

c $\frac{18\sqrt{t^5}}{5}\mathbf{i} - \frac{8t^3}{3}\mathbf{j} + \mathbf{c}$

d $(2e^{-t} + 6t)\mathbf{i} + \left(\frac{3t^2}{2} + 7t\right)\mathbf{j} + \mathbf{c}$

e $\frac{-\cos(3t)}{3}\mathbf{i} + \sin(t)\mathbf{j} + \mathbf{c}$

f $\frac{4\sin(5t)}{5}\mathbf{i} + \frac{8e^{3t}}{3}\mathbf{j} + \mathbf{c}$

2 **a** $\mathbf{r}(t) = t\mathbf{i} + [(t-1)^3 + 1]\mathbf{j}$

b $\mathbf{r}(t) = 20t\mathbf{i} + (15t - 5t^2 + 4)\mathbf{j}$

c $\mathbf{r}(t) = 2\cos(t)\mathbf{i} + [3\sin(t) + 2]\mathbf{j}$

d $\mathbf{r}(t) = e^t\mathbf{i} + (e^{-t} + 1)\mathbf{j}$

e $\mathbf{r}(t) = \frac{t+1}{2}\mathbf{i} + t^2\mathbf{j}$

f $\mathbf{r}(t) = (e^t + 2)\mathbf{i} + \frac{2}{9}e^{2t}\mathbf{j}$

3 **a** $\mathbf{r}(t) = \left(\frac{3t^2}{2} + 2\right)\mathbf{i} + (5t - 3)\mathbf{j}$

b $\mathbf{r}(t) = -(3\cos(t) + 1)\mathbf{i} + \left[\frac{1}{2}\sin(2t) + 4\right]\mathbf{j}$

c $\mathbf{r}(t) = \frac{4 + e^{3t}}{3}\mathbf{i} + 5(3 - e^{-t})\mathbf{j}$

d $\mathbf{r}(t) = \frac{1}{2}(t+2)\mathbf{i} + (t^2 - 3)\mathbf{j}$

4 **a** $8\mathbf{i} - 2\mathbf{j}$

b $\frac{2}{3}\mathbf{j}$

c $\mathbf{i} - \mathbf{j}$

d $\frac{-\sqrt{3}}{2}\mathbf{j}$

e $2\mathbf{i} - \frac{-(e^5 - 1)}{5}\mathbf{j}$

f $\frac{1 - e^{-2}}{2}\mathbf{i} + 2(e-1)\mathbf{j}$

5 $\frac{1}{4}(e^8 - 1)\mathbf{i} - \frac{2}{3}(\sqrt{3^3} - 1)\mathbf{j}$

6 $\frac{1}{3}\mathbf{i} + \frac{1}{3}\mathbf{j}$

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